EGMT 1520: BUILDING TRUTH FROM SCRATCH (EMPIRICAL & SCIENTIFIC ENGAGEMENT)

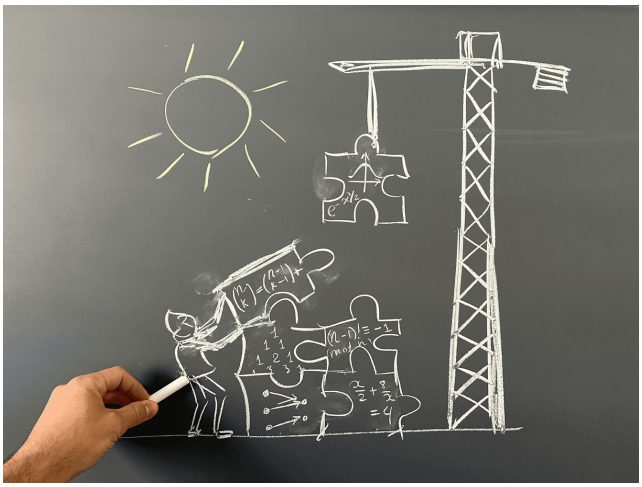
LEONID PETROV
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1. *How do we know a claim is true?*

This course is a hands-on workshop in making and testing arguments in the context of mathematics. We will generate conjectures from examples, search for counterexamples, and turn ideas into precise statements and proofs. Through problem-solving sessions and presentations, you'll practice evaluating arguments, giving and receiving constructive feedback, and communicating clearly in writing and speech. By experiencing mathematics as a creative process — where patterns suggest conjectures and logical reasoning turns intuition into conviction — you'll develop a practical sense for what counts as evidence in mathematics and theoretical sciences, and how to build reliable conclusions.

By the end of the course, you will be able to

- **Define and delimit what constitutes valid mathematical evidence** by distinguishing between examples, counterexamples, conjectures, and formal proofs, while recognizing the limitations of empirical observations.
- **Develop a framework for discerning different types of mathematical knowledge** by exploring how empirical evidence, abstract reasoning, and logical structure work together to shape mathematical understanding.
- **Formulate and communicate mathematical reasoning** by translating intuitive insights into precise statements, evaluating the soundness of arguments, and engaging in constructive dialogue to identify and resolve reasoning gaps.
- **Reflect on the nature of mathematical truth** by examining personal assumptions about certainty, analyzing when and why certain arguments are conclusive, and articulating how purely empirical approaches can both inform and limit our understanding of complex phenomena.



Date: Compiled on Monday 19th October, 2026, 11:26. An up to date syllabus is always at this link. An accessible HTML version is at this link.

2. *Contact & Logistics*

Instructor: Leonid Petrov
Email: petrov@virginia.edu
Website: https://lpetrov.cc
Office: 209 Kerchof Hall
Office hours:
Mon 2–3pm, Wed 11am–12pm
or by appointment (email me; you can make as many appointments as you need)

Course Info: EGMT 1520-206 (19629) — Empirical Engagement.

Weekly rhythm: Class meets **Mondays and Wednesdays 3:30PM – 4:45PM** in **Gilmer Hall 247**.

- **Dates:** October 19, 2026 – December 7, 2026.
- **No Class:** November 25 (Thanksgiving break).
- **All 14 class meetings, by week:**
  - Week 1: Mon Oct 19, Wed Oct 21
  - Week 2: Mon Oct 26, Wed Oct 28
  - Week 3: Mon Nov 2, Wed Nov 4
  - Week 4: Mon Nov 9, Wed Nov 11
  - Week 5: Mon Nov 16, Wed Nov 18
  - Week 6: Mon Nov 23 — *special meeting* — *Math Casino* (no class Wed — Thanksgiving)
  - Week 7: Mon Nov 30, Wed Dec 2
  - Week 8: Mon Dec 7

Course materials: There are no required textbooks for this course. All materials will be handed out in class and later posted on Canvas. The Commonplace Book is an integral part of the course for homework assignments and reflections — I recommend you bring it to each class session. Manipulatives are provided in class.

Canvas and Gradescope: Problem sets and weekly writing assignments will be posted on Canvas. You submit your written work to Gradescope (linked from Canvas).

Accessibility: In class, this course is pen-and-paper: printed handouts, the board, the Commonplace Book, and manipulatives. The only digital parts are Canvas (problem sets, assignments, and announcements) and Gradescope (submitting photos or scans of your written work). If you have trouble reading, viewing, or using any of these materials, printed or digital, please let me know right away so we can work through this together.

3. *How we work together in class*

This is a hands-on, pen-and-paper course. Each topic moves from intuition to rigor, usually over one or two class meetings; the pace follows the class, not a fixed schedule.

3.1. **Class activities.** A topic typically moves through the activities below. Their order, length, and mix vary from meeting to meeting, depending on the topic and on how the class is progressing.

- **Primer:** a short introduction to the core idea, so you have the tools you need before you start solving.
- **Problem solving:** you work on a problem set in your home subgroup. Problems are roughly ordered by difficulty (Easy → Medium → Hard). I circulate, ask questions, and may invite nearby peers to listen in and challenge your reasoning.
- **Presentations:** a student or a subgroup presents a solution, complete or partial, on the board. The rest of the class acts as "Skeptics": asking clarifying questions, testing the argument, and pointing out gaps. Some presentations serve as model solutions that show what good structure, clarity, and logical flow look like.
- **Acceptance:** your subgroup aims to have its solutions "accepted" by the instructor. A problem is "accepted" when your subgroup can answer follow-ups and defend key steps without unresolved gaps.

3.2. **Home subgroups (fixed).** On Week 2 we form "home subgroups" for in-class collaboration: δ, θ, ζ, ρ, λ, ϕ.[1] Subgroups remain stable throughout the course.

3.3. **Team roles (rotate each meeting).** In subgroups of 5–6, the following sample roles work well:

- Explainer: articulates the current approach and restates the problem in your own words.
- Skeptic: presses on assumptions, looks for gaps, and frames precise questions.
- Counterexample Hunter: designs and tests examples/edge cases to probe the claim.
- Recorder: maintains a clean write-up of the steps in the problem.
- Verifier: checks computations and logic, and ensures steps follow from the stated problem.
- Connector: ensures that all subgroup members understand and can explain the solution.

You may keep the roles informal or explicitly assign them in your subgroup, but *make sure you rotate them at least each week, if not more often.*

3.4. **Commonplace book.** The Commonplace Book is used for your weekly writing assignments and reflections, which you submit as photos to Gradescope (linked from Canvas). It is helpful to bring it to class each time.

3.5. **Topic.** The weekly math topic is not announced in advance — we need everyone to bring their curiosity and creativity to the class for a shared joy of discovering new ideas.

3.6. **Devices.** Please keep devices in your bag unless you have an accommodation that requires otherwise. This is a pen-and-paper course.

4. *Commonplace Book weekly assignments*

Each week you complete two assignments by hand in your Commonplace Book: a weekly reflection and one problem write-up. Submit clear photos or a single PDF scan of the pages to Gradescope (linked from Canvas) by **Sunday 10:00 pm**, each to its own assignment.

(1) **Weekly reflection (300 words / 1 page limit)** — question changes every week. Reflections are graded on a 10-point scale for thoughtfulness and engagement with ideas from class and your subgroup. Examples:
   (a) **What is the hardest mathematical fact I have ever seen and understood in my life?** (*first week assignment*). Describe your mathematical journey in your own words (not just list math classes you took). What was the hardest mathematical thing you have ever seen and understood? *This assignment will help me form balanced home subgroups starting in the second week.*
   (b) **What counted as evidence for me?** Name one math experience from this week and list *exactly what* made it convincing *to you* (e.g., a minimal example, a failed counterexample, a definition that removed ambiguity, an auxiliary statement that closed a gap). End with one thing that would change your mind about this experience being convincing.
   (c) **Reflect on the nature of mathematical truth.** What does it mean for a mathematical claim to be true? How is it different from a scientific claim, like "water boils at 100 degrees Celsius"? What is the role of examples, counterexamples, and proofs in establishing mathematical truth?
   (d) **Assumption audit.** Pick one problem from the problem set, and list all assumptions you used in your solution (not just those stated in the problem). For each, mark: *needed* / *not needed* / *uncertain*. Then try to *drop one*: either give a tiny counterexample that shows it *was* needed, or a brief note explaining why the argument still goes through.

(2) **Problem write-up (1–2 pages).** Choose exactly one problem marked ✏ from the week's problem sets, and write a clear, self-contained solution for a classmate from another subgroup who has the handout but did not hear your table discussion. Your write-up must include:

[1] Math and science use Greek symbols very often. The names of the subgroups are *delta*, *theta*, *zeta*, *rho*, *lambda*, and *phi*.

- **Problem restatement:** the problem in your own words, clarifying what is given and what must be shown.
- **Definitions and setup:** notation and any terms you rely on.
- **Argument:** your proof or solution, in logical sequence.
- **Clear conclusion:** your final conclusion, stated explicitly.

Write-ups are graded *Meets* (2) / *Revise* (1) / *Missing* (0). A *Meets* write-up is correct, logically complete, and easy to follow.

Week 1 exception: the first reflection, example (a) above, is due **Thursday, October 22, 10:00 pm**, so that I can form the home subgroups before Week 2. The first write-up is due Sunday, October 25.

5. *Engaging Grounds*

This component is an integral part of the Engagements program. The Quarter 2 tasks are on pp. 209–213 of your Commonplace Book:

(1) **Discover your values** (pp. 209–210; complete it before you meet with your advisor). List the 5–10 things that matter most to you. Rank them by how much you care about them, and again by how much time and attention you give them. How closely do the two rankings align, and what would you change? Which majors, courses, career paths, or activities fit your values?

(2) **Meet with your advisor** (p. 211). Everyone is required to meet with their advisor during Quarter 2 to prepare for Spring enrollment. Enrollment begins by appointment on November 2, so meet before your own enrollment appointment. Come prepared to answer: Which courses do you most want to take in the Spring? What have you learned this semester that will change how you choose courses? Record your advisor's name, the date of the meeting, and your notes from it.

(3) **Attend an academic event** (p. 212). Attend one Academic Event from the Engagements Calendar and respond to the prompts: Which event did you attend? Where and when was it? What academic or personal reflections prompted you to select it? What questions did it provoke? Would you recommend it to another student? Why or why not?

(4) **Map your path** (p. 213). On the map of Grounds on pp. 222–223, mark the places you visited for this quarter's Engaging Grounds tasks (your advisor meeting, the academic event, and a quiet place to reflect), along with a friend's dorm and your favorite place to study. Use your own symbols, drawings, or stickers.

Completion must be documented in the Commonplace book, and photos of the pages must be submitted to Gradescope (linked from Canvas) by **Wednesday, December 9, 10:00 pm**. The Engaging Grounds component is worth 10% of the final grade.

6. *Grading*

- **In-class work & explanations (50%):** Assessed on:
  - **Clarity** of explanations at the table.
  - **Responsiveness** to questions and proposed counterexamples (can you repair, refine, or retract a claim when pressed?).
  - **Equitable participation** within the subgroup (roles rotate; multiple voices contribute to each explanation).

  Attendance is required to earn credit. A problem is "accepted" when your subgroup can answer follow-ups and defend key steps without unresolved gaps.
- **Commonplace notebook (40%):** Weekly writing assignments, submitted to Gradescope (linked from Canvas) by Sunday 10:00 pm as photos or a PDF scan. Problem write-ups are graded on the following scale: *Meets* (2, full credit), *Revise* (1; resubmission within 72 hours yields full credit), or *Missing* (0). Weekly reflections are graded on a 10-point scale. Your lowest problem write-up and your lowest reflection are both dropped automatically.

- **Engaging Grounds (10%):** Complete the assigned tasks in the Commonplace book, and submit photos of the pages to Gradescope (linked from Canvas) by Wednesday, December 9, 10:00 pm.

The percentage grade is *not rounded up*, and is calculated according to the following scale:

100+	A+	83–86.99	B	67–69.99	D+
93–99.9	A	80–82.99	B-	63–66.99	D
90–92.99	A-	77–79.99	C+	60–62.99	D-
87–89.99	B+	73–76.99	C	< 60	F
70–72.99	C-				

7. *Policies*

7.1. **Late work.** Each weekly notebook assignment is due on Sunday at 10pm.[2]

You have one no-questions-asked Grace Week: submit any one weekly notebook by Wednesday 10 pm with no penalty. Beyond that, late assignments are not accepted. If you have special needs or an emergency, please let me know as soon as possible.

7.2. **Honor Code.** The University of Virginia Honor Code applies to this course and is taken seriously. Any Honor Code violations will be referred to the Honor Committee.

7.3. **Devices.** This is a pen-and-paper course. Please keep all devices in your bag during class activities, unless you have an accommodation that directs otherwise.

7.4. **Use of AI.**

- **Allowed for learning**: asking conceptual questions; generating small practice examples; checking ideas (not text) for plausibility.
  For learning with LLMs, here is an example `prompt` that might be helpful.
- **Allowed with citation**: grammar/clarity edits to your own draft (light copy-editing only). If used, add a one-line note at the end: AI-assisted for copy-editing only.
- **Not allowed**: drafting any part of the submitted solution or reflections; step-by-step proofs; problem-specific hints beyond what's provided in class; rewriting your math into polished prose.
- **Which tool to use**: any AI tool is fine for the allowed uses above. UVA ITS lists the AI tools supported at UVA; Microsoft Copilot Chat and Google Gemini are available to all students at no cost. Signed in with your UVA account, what you type stays within UVA and is not used to train AI models. You are never required to create an account on an outside service for this course.
  University-wide guidance is in `Guidance for Faculty & Students`, and there is a student-facing `AI for Academic Excellence toolkit`. Where the rules above are stricter than the University guidance, the rules above apply.

7.5. **Attendance.** Consistent attendance and thoughtful in-class participation are absolutely essential for your successful completion of Engagements courses. Since each course is only 14 classes, each absence is significant. That said, there are situations in which you may need to miss a class meeting. In those situations, Engagements professors are asked to observe the following policy:

- (excused absences) Authorized university activities (such as university-sponsored athletic events) and the observance of religious holidays are considered excused absences. However, you must submit a written request detailing the dates of these absences directly to me by email by **October 27**. If these types of absences would result in you missing more than 2 of the required class meetings, you may need to switch to a different course with fewer scheduling conflicts.
- (other causes of absences) We all get sick and we all have experienced unexpected family and personal emergencies. If you must miss class for any reason, you should notify me in advance, when possible, and you are responsible for making up any missed work. In line with College policy, professors are under no obligation to offer alternative assignments or to organize and facilitate make-up work: students must take the initiative.

[2] With a small forgiveness window for technical issues.

After the second absence, any subsequent unexcused absence will incur an automatic one letter-grade (or 10 out of 100 points) penalty. For example:

- A student with "A" level work who misses a total of 3 classes would receive a "B" letter grade.
- A student with "A" level work who misses a total of 4 classes would receive a "C" letter grade.

Important Reminder. If you are experiencing significant and pressing personal circumstances, particularly if those circumstances interfere with your ability to attend class and complete coursework, you should contact your academic advisor to talk through the situation as soon as possible. Help is available, and these situations can almost always be worked out — but you need to let me know that there's a problem.

7.6. **Special needs.** All students with special needs requiring accommodations should present the appropriate paperwork from the Student Disability Access Center (SDAC). It is the student's responsibility to present this paperwork in a timely fashion and to follow up with the instructor about the accommodations being offered.